PAYETTE SUPERVISOR'S OFFICE

PAYETTE NATIONAL FOREST

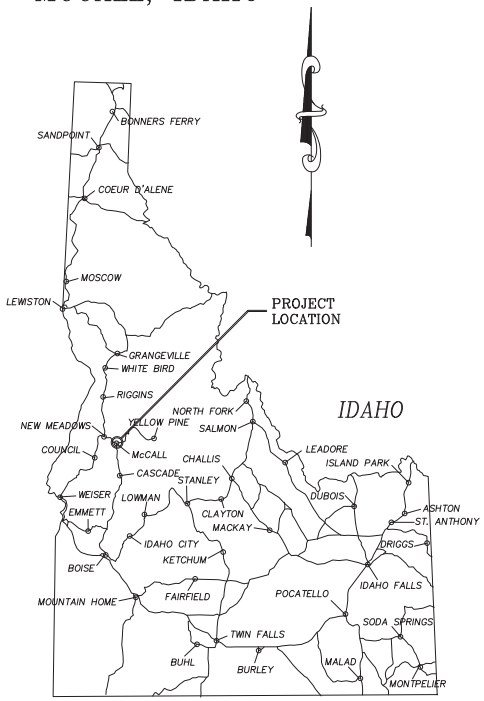
MCCALL, IDAHO

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RECOMMENDED

Will Perry — 6-19-13

FOREST ENGINEER — DATE

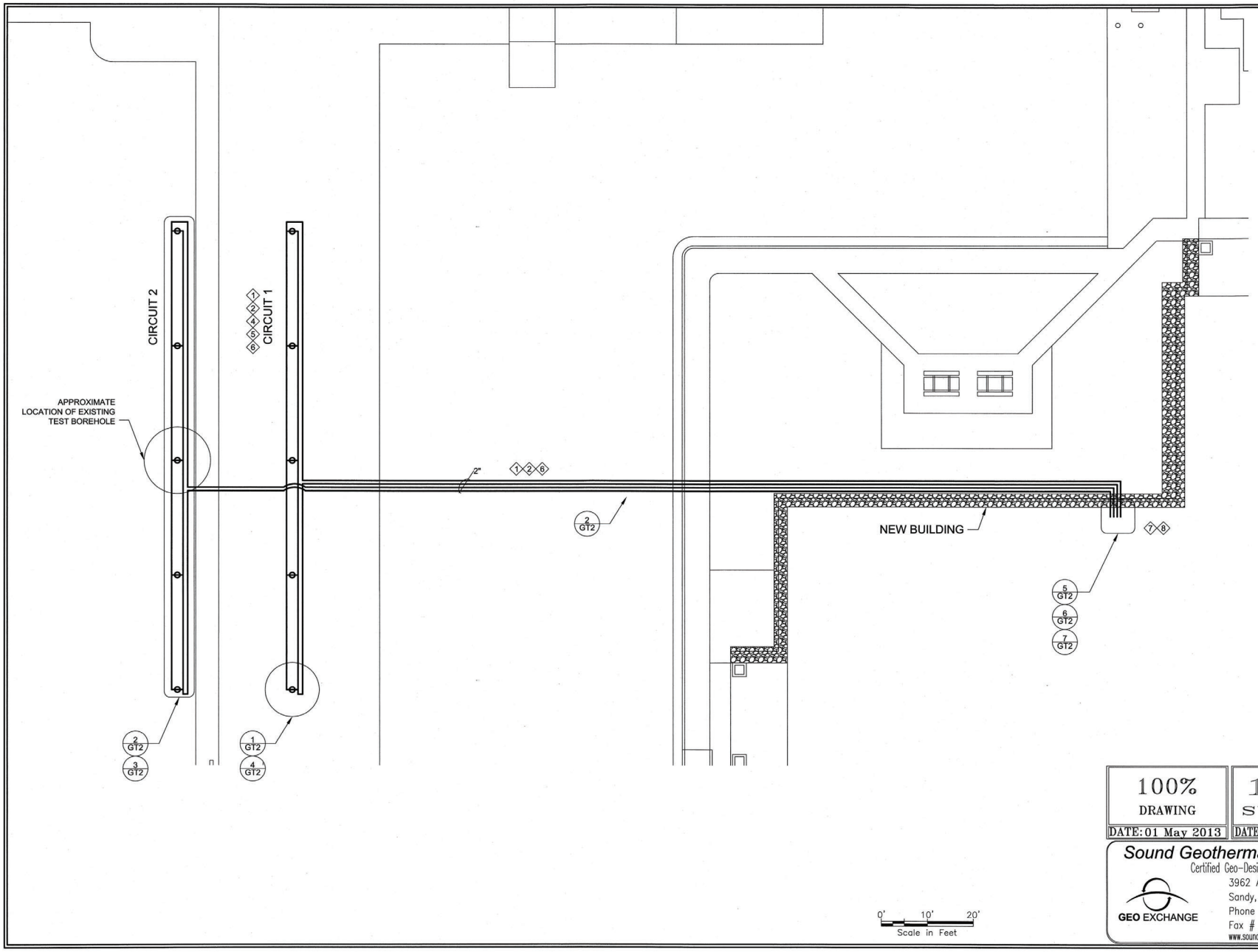
APPROVED

Keith Lannom — 6-19-13

FOREST SUPERVISOR — DATE

Gina Freel — 6-27-13

DIRECTOR, REGION 4 ENGINEERING — DATE



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FOREST SERVICE
Intermountain Region 4 Engineering
PAYETTE NATIONAL FOREST

DESIGN By: M. SMITH CHECK: D. C. SMITH
DRAWING By: M. SMITH CHECK: D. C. SMITH
APPROVED: DATE 6.27.13

PAYETTE SUPERVISOR'S OFFICE

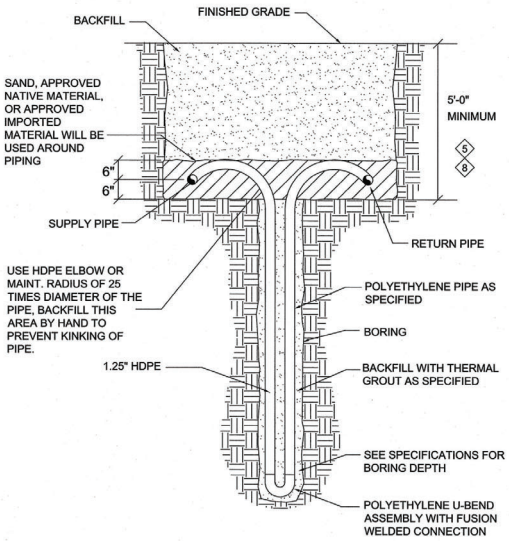
GEOTHERMAL SITE PLAN

LICENSED PROFESSIONAL ENGINEER NO. 5707707 MATTHEW O. MARSHALL STATE OF UTAH
5/10/13

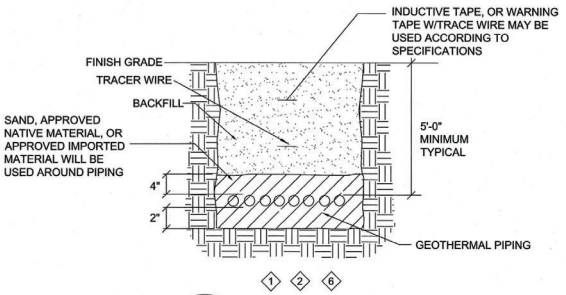
100% DRAWING	100% SUBMITTAL
DATE: 01 May 2013	DATE: 01 May 2013

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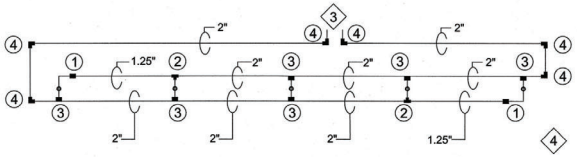
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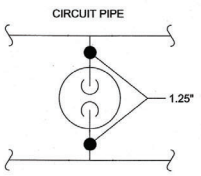
1 BORE HOLE DETAIL.
Scale: NTS (TYPICAL)



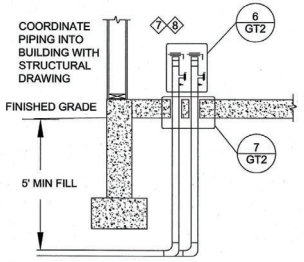
2 TRENCH DETAIL.
Scale: NTS (TYPICAL)



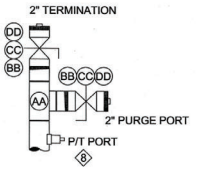
3 CIRCUIT PIPE DETAIL.
Scale: NTS (TYPICAL)



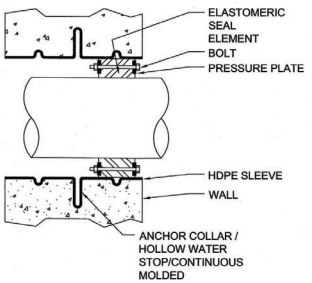
4 BORE HOLE CONNECTION.
Scale: NTS (TYPICAL)



5 MECH. ROOM PENETRATION.
Scale: NTS (TYPICAL)



6 MECHANICAL ROOM TERMINATION DETAIL.
Scale: NTS (TYPICAL)



7 FLOOR/WALL PENETRATION DETAIL.
Scale: NTS

HDPE PIPE SCHEDULE

PIPE SIZE	SDR RATING
1.25" HDPE U-BENDS - 610'	11
1.25" HDPE	11
2" HDPE	11

HEADER FITTING SCHEDULE

(#)	SIZE
1	1.25" HDPE COUPLING
2	2" x 1.25" x 1.25" HDPE TEE
3	2" x 2" x 1.25" HDPE TEE
4	2" HDPE ELBOW

MECHANICAL ROOM FITTING SCHEDULE

(**)	SIZE
AA	2" x 2" x 2" HDPE TEE
BB	2" HDPE x MPT TRANSITION
CC	2" FULL PORT BALL VALVE
DD	2" MPT PLUG

KEYED NOTES (ALL MS SHEETS)

1. RUN SUPPLY AND RETURN PIPING IN COMMON TRENCH. INSTALL PIPING SIMILAR TO ARRANGEMENT SHOWN IN TRENCH DETAIL.
2. RUN SUBFIELD PIPING MINIMUM 60" BELOW GRADE EXCEPT WHERE NOTED. TRENCHES MUST MEET OSHA REQUIREMENTS.
3. GSS AND GSR TO MECHANICAL ROOM. SEE CONTINUATION IN DETAIL 5 ON MS501.
4. BORE HOLE SPACING TO BE 25'-0" O.C. EACH DIRECTION, EXCEPT WHERE NOTED.
5. BORE HOLE DEPTH IS 400', SEE SPECIFICATIONS AND DETAILS, TYPICAL.
6. FOR COMPACTION REQUIREMENTS AND TYPE OF BEDDING MATERIAL REFER TO APPROPRIATE EARTH WORK SECTION OF SPECIFICATIONS.
7. MECHANICAL ENGINEER'S DRAWINGS WILL CONTROL FINAL LAY-OUT.
8. INSTALL PT PORT PRIOR TO PURGE PORT.

GENERAL NOTES (ALL MS SHEETS)

1. DEPTH OF THE HORIZONTAL PIPING SHALL BE INCREASED AS NEEDED TO INSURE MINIMUM CODE SEPARATION FROM UTILITIES.
2. INDIVIDUAL BOREHOLES MAY BE MOVED +/- 4'-5' TO AVOID UTILITIES. RELOCATED BOREHOLES MUST BE SPECIFICALLY IDENTIFIED ON AS-BUILT DRAWINGS.
3. ALL EXCAVATION SHALL COMPLY WITH OSHA REGULATIONS.

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U. S. Department of Agriculture
FOREST SERVICE Engineering
Intermountain Region 4
PAYETTE NATIONAL FOREST

	BY:	CHECK:	DATE
DESIGN	M. SMITH	D. C. SMITH	4-27-13
DRAWING	M. SMITH	D. C. SMITH	
APPROVED: Jim Tait, DIRECTOR ENGINEERING			

PAYETTE SUPERVISOR'S OFFICE

GEOTHERMAL DETAILS

100% DRAWING	100% SUBMITTAL
DATE: 01 May 2013	DATE: 01 May 2013

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LICENSED PROFESSIONAL ENGINEER NO. 5707707 MATTHEW O. MARSHALL STATE OF UTAH 5/3/13

PROJECT NO. 5586 | DRAWING MS501 | SHEET 74 of 88

PIPING AND EQUIPMENT INSULATION SCHEDULE

SYMBOL	SYSTEM	EXPOSED	ABOVE CEILING	EXTERIOR EXPOSED	THICKNESS	INDOOR FIELD APPLIED JACKET	OUTDOOR FIELD APPLIED JACKET
GLS / GLR	GEOTHERMAL LOOP	PRE-FORMED MINERAL FIBER	--	--	SEE THICKNESS TABLE BELOW	PVC	--

MINIMUM PIPE INSULATION THICKNESS TABLE

FLUID DESIGN OPERATING TEMP. RANGE (°F)	INSULATION CONDUCTIVITY		NOMINAL PIPE OR TUBE SIZE (in.)				
	CONDUCTIVITY Btu-in/(h-ft²-°F)	MEAN RATING TEMP. °F	<1"	1" TO <1 1/2"	1 1/2" TO <4"	4" TO <8"	≥ 8"
COOLING SYSTEMS (CHILLED WATER, BRINE, AND REFRIGERANT)							
40-60	0.22-0.28	100	0.5	0.5	1.0	1.0	1.0
<40	0.22-0.28	100	0.5	1.0	1.0	1.0	1.5

NOTES:
1. ANY PIPE ON THIS SCHEDULE WITHIN THE FACILITY TO BE INSULATED AS DEFINED.
2. PIPING INSULATION IS NOT REQUIRED BETWEEN THE CONTROL VALVE AND COIL ON RUN-OUTS WHEN THE CONTROL VALVE IS LOCATED WITHIN 4 FT OF THE COIL AND THE PIPE SIZE IS 1 IN. OR LESS.
3. THESE THICKNESSES ARE BASED ON ENERGY EFFICIENCY CONSIDERATIONS ONLY. ISSUES SUCH AS WATER VAPOR PERMEABILITY OR SURFACE CONDENSATION SOMETIMES REQUIRE VAPOR RETARDERS OR ADDITIONAL INSULATION.

DUCT INSULATION SCHEDULE

DUCT TYPE	DUCT LOCATION	INSULATION R-VALUE	REMARKS
SUPPLY, RETURN	NOT WITHIN CONDITIONED SPACE : ON EXTERIOR OF BUILDING, ON ROOF, IN ATTIC, IN ENCLOSED CEILING SPACE, IN WALLS, IN GARAGE, IN CRAWL SPACES	R-6	
OUTSIDE AIR INTAKE	WITHIN CONDITIONED SPACE	R-6	
SUPPLY, RETURN, OUTSIDE AIR INTAKE	NOT WITHIN CONDITIONED SPACE	R-3.5	
SUPPLY W/ SUPPLY AIR TEMP. <55°F OR >105°F	WITHIN CONDITIONED SPACE	R-3.5	

NOTE:

REQUIREMENTS APPLIED TO THE DUCT TYPE LISTED, WHETHER HEATED OR MECHANICALLY COOLED. MECHANICALLY COOLED DUCTS REQUIRING INSULATION SHALL HAVE A VAPOR RETARDER, WITH A PERM RATING NOT GREATER THAN 0.5 AND ALL JOINTS SEALED.

INSULATION TYPES: MINIMUM DENSITIES AND OUT OF PACKAGE THICKNESS. NOMINAL R-VALUES ARE FOR THE INSULATION AS INSTALLED AND DO NOT INCLUDE AIR FILM RESISTANCE.

R-3.5 1.0 INCH 1.5 TO 3.0 lb/cu.ft. DUCT LINER, MINERAL OR GLASS FIBER BLANKET OR EQUIVALENT TO PROVIDE AN INSTALLED TOTAL THERMAL RESISTANCE OF AT LEAST R-3.5.

R-6 3.0 INCH 0.75 lb/cu.ft. MINERAL OR GLASS FIBER BLANKET, 2.0 INCH 1.5 TO 3.0 lb/cu.ft.DUCT LINER, MINERAL OR GLASS FIBER BLANKET, 2.0 INCH 3.0 TO 7.0 lb/cu.ft. MINERAL OR GLASS FIBER BOARD OR EQUIVALENT TO PROVIDE AN INSTALLED TOTAL THERMAL RESISTANCE OF AT LEAST R-6.

BRANCH SELECTOR BOX SCHEDULE

MARK	MANUFACTURER & MODEL	DESCRIPTION	MAX. CAPACITY (MBH)	CONNECTABLE UNITS	ELECTRICAL HP	ELECTRICAL VOLTS/ PHASE	DIMENSIONS HxWxD (IN)	WEIGHT (LBS)	REMARKS
BSB 1	DAIKIN BSVQ36PVJU	REFRIGERANT CHANGE OVER SELECTOR	36	MAX. 5	--	208/1	8x16x13	26	MOUNT ABOVE CEILING
BSB 2	DAIKIN BSVQ60PVJU	REFRIGERANT CHANGE OVER SELECTOR	60	MAX. 8	--	208/1	8x16x13	26	MOUNT ABOVE CEILING

CENTRIFUGAL PUMP SCHEDULE

MARK	MANUFACTURER & MODEL	SYSTEM SERVED	TYPE	GPM	HEAD (FT)	FLUID	FLUID TEMP	IMPELLER DIA (IN)	MOTOR RPM	MOTOR HP	MOTOR VOLT/ PHASE	REMARKS
P 1	BELL & GOSSETT SERIES 80 1 1/2x1 1/2x7B	GEOTHERMAL LOOP	INLINE CENTRIF.	54	50	25% P.G.	35	7	1,750	1.5	208 / 3	
P 2	BELL & GOSSETT SERIES 80 1 1/2x1 1/2x7B	GEOTHERMAL LOOP	INLINE CENTRIF.	54	50	25% P.G.	35	7	1,750	1.5	208 / 3	

AIR SEPARATOR SCHEDULE

MARK	MANUFACTURER & MODEL	SYSTEM SERVED	CONN. SIZE (IN)	MAX GPM	DESIGN GPM	PD FT WC	DIMENSIONS DIA (IN)	DIMENSIONS H (IN)	DIMENSIONS FLG TO FLG (IN)	DIMENSIONS DRY WT (LBS)	STRAINER FREE AREA (SQ IN)	REMARKS
AS 1	SPIROTHERM SPIROVENT VSR-250-MT	GEOTHERMAL LOOP	2 1/2	70	54	1	6.3	19	15.7	28	N/A	

DIRECT EXPANSION DUCT FREE SPLIT SYSTEM SCHEDULE

MARK	AREA SERVED	NOM. TONS	SEER	REFRIG	TOTAL MBH COOL	INDOOR UNIT MANUFACTURER & MODEL	INDOOR UNIT CFM	INDOOR UNIT AMPS	INDOOR UNIT DIMENSIONS HxWxD (IN)	INDOOR UNIT WEIGHT (LBS)	OUTDOOR UNIT MANUFACTURER & MODEL	OUTDOOR UNIT VOLT/ PHASE	OUTDOOR UNIT MCA	OUTDOOR UNIT MOCP	OUTDOOR UNIT DIMENSIONS HxWxD (IN)	OUTDOOR UNIT WEIGHT (LBS)	REMARKS
DX 1	COMM ROOM	3.5	16	R-410A	30.9	DAIKIN CASSETTE FCQ42PAVJU	1,220	1.5	12x33x33	67	DAIKIN SKYAIR RZR42PVJU	208/1	27	30	53x36x13	283	LOW AMBIENT CONTROL, CRANKCASE HEATER REMOTE CONTROL THERMOSTAT, COOLING ONLY SYSTEM.

DUCT HEATER SCHEDULE

TAG	MAKE/MODEL	CFM	WIDTH	HEIGHT	kW	VOLTAGE	ACCESSORIES
DH-1	INDEECO QUA	1200	20 IN	12 IN	12	208/3/60	A-G

EXHAUST FAN SCHEDULE

MARK	MANUFACTURER & MODEL	DESCRIPTION	AREA SERVED	CFM	STATIC PRESS. (IN WG)	RPM	FAN TYPE	SOUND RATING (DBA)	ELECTRICAL HP	ELECTRICAL VOLTS/ PHASE	WEIGHT (LBS)	REMARKS
EF 1	COOK 89-SQNB	INLINE CENTRIF.	RESTROOMS, JANITOR RM.	610	0.5	1,927	CENTRIFUGAL	64	1/4	120 / 1	80	1, 7

REMARKS:
1. ALL CONDITIONS ARE FOR AN ELEVATION OF 5,000 FEET ABOVE SEA LEVEL.
2. FAN TO BE SPARK PROOF CONSTRUCTION.
3. FURNISH FAN WITH FACTORY MOUNTED DISCONNECT SWITCH.
4. FAN TO BE AMCA B SPARK-PROOF CONSTRUCTION.
5. FACTORY MANUFACTURED AND SIZED OUTLET FLANGE AND HOUSED VIBRATION ISOLATORS TO BE PROVIDED WITH FAN.
6. FAN INLET TO BE PROVIDED WITH COMPANION FLANGE.
7. PROVIDE UNIT WITH FACTORY MOUNTED GRAVITY BACKDRAFT DAMPER.
8. FAN TO BE VARIABLE SPEED TO MAINTAIN NEGATIVE PRESSURE.
9. FAN TO BE VARIABLE SPEED.

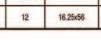
WATER SOURCE HEAT RECOVERY UNIT SCHEDULE

MARK	MANUFACTURER & MODEL	NOM. TONS	GPM	HEATING CAPACITY (MBH)	COOLING CAPACITY (MBH)	HTG. COP	CLG. EER	REFRIG. TYPE	ELECTRICAL VOLTS/ PHASE	ELECTRICAL MCA	ELECTRICAL MFS	PIPE CONNECTIONS (IN) GAS	PIPE CONNECTIONS (IN) LIQUID	PIPE CONNECTIONS (IN) H/L P.C.	PIPE CONNECTIONS (IN) H/L EQ. CONN.	REMARKS
HRU 1	DAIKIN AC RWEYQ252PTJU-HR	21	197	197	245	4.5	12.8	R-410A	208 / 3	22.4/3	40x3	SEE SCHEM.	SEE SCHEM.	SEE SCHEM.	SEE SCHEM.	UNIT IS A COMBINATION OF THREE 7-TON UNITS

REGISTER, GRILLE & DIFFUSER SCHEDULE

MARK	MANUFACTURER & MODEL	DESCRIPTION	MATERIAL	FINISH	FRAME	DAMPER	MAX. NC	MAX. CFM	MAX. APD	NECK SIZE	REMARKS
A	TITUS TDV-AA	CEILING LOUVERED SUPPLY DIFFUSER	ALUMINUM	WHITE	LAY-IN	--	30	140	0.14	6x6"	
B	TITUS TDV-AA	CEILING LOUVERED SUPPLY DIFFUSER	ALUMINUM	WHITE	SURFACE MOUNT	--	30	140	0.14	6x6"	
C	TITUS TDV-AA	CEILING LOUVERED SUPPLY DIFFUSER	ALUMINUM	WHITE	LAY-IN	--	30	325	0.14	9x9"	
D	TITUS TDV-AA	CEILING LOUVERED SUPPLY DIFFUSER	ALUMINUM	WHITE	SURFACE MOUNT	--	30	325	0.14	9x9"	
E	TITUS TDV-AA	CEILING LOUVERED SUPPLY DIFFUSER	ALUMINUM	WHITE	LAY-IN	--	30	525	0.14	12x12"	
F	TITUS 350R	CEILING LOUVERED RETURN GRILLE	ALUMINUM	WHITE	LAY-IN	--	--	1,100	0.05	22x22"	
G	TITUS 350R	WALL/CEILING LOUVERED RET. / EXHAUST GRILLE	ALUMINUM	WHITE	SURFACE MOUNT	--	16	220	0.08	6x6"	
H	TITUS 350R	WALL/CEILING LOUVERED RET. / EXHAUST GRILLE	ALUMINUM	WHITE	SURFACE MOUNT	--	19	350	0.08	10x10"	
J	TITUS 350R	WALL/CEILING LOUVERED RET. / EXHAUST GRILLE	ALUMINUM	WHITE	SURFACE MOUNT	--	19	520	0.08	12x12"	
K	TITUS 350R	WALL/CEILING LOUVERED RET. / EXHAUST GRILLE	ALUMINUM	WHITE	SURFACE MOUNT	--	19	700	0.08	14x14"	
L	TITUS 350R	WALL/CEILING LOUVERED RET. / EXHAUST GRILLE	ALUMINUM	WHITE	SURFACE MOUNT	--	21	920	0.08	16x16"	
M	TITUS 300SR-HD	SUPPLY GRILLE	STEEL	WHITE	DUCT MOUNT	O.B.D.	20	180	0.10	8x6"	
N	TITUS 300SR-HD	SUPPLY GRILLE	STEEL	WHITE	DUCT MOUNT	O.B.D.	23	255	0.13	8x8"	

NOTE: ALL DIFFUSERS AND GRILLES LISTED IN SCHEDULE MAY NOT APPEAR ON DRAWINGS.



SYMBOL IDENTIFICATION DETAIL

HEAT PUMP SCHEDULE

MARK	MANUFACTURER & MODEL	AREA SERVED	NOM. TONS	IEER	REFRIG	TOTAL MBH COOL	TOTAL MBH HEAT	PIPE CONN (IN)	VOLTS/ PHASE	MCA	MOCP	DIMENSIONS HxWxD (IN)	WEIGHT (LBS)	REMARKS
HP 1	DAIKIN RXQ36PXBTJ	OUTSIDE AIR PROCESSING	8	23	R-410A	92	103	7/8 / 3/8	208/3	43	50	66x49x30	633	

EXPANSION TANK SCHEDULE

MARK	MANUFACTURER & MODEL	SYSTEM SERVED	TYPE	TEMP (°F)	FLUID	TANK VOL GAL	ACCEPT. VOL GAL	PRE-CHARGE PSI	DIMENSIONS DIAxH (IN)	WT (LBS)	REMARKS
ET 1	BELL & GOSSETT D-60V	GEOTHERMAL LOOP	DIAPHRAGM	85	25% P.G.	44.4	22.6	12	16.25x56	146	

MECHANICAL LEGEND

SYMBOL	DESCRIPTION
GLS	GEOTHERMAL LOOP SUPPLY
GLR	GEOTHERMAL LOOP RETURN
OR	MANUAL VOLUME CONTROL DAMPER
A	AUTOMATIC TEMP. CONTROL DAMPER
	FIRE DAMPER (FD)
OR	EXHAUST AIR DUCT - UP / DOWN
OR	RETURN/OUTSIDE AIR DUCT - UP / DOWN
OR	SUPPLY AIR DUCT - UP / DOWN
	VAV BOX
	UNIT HEATER
	MITERED ELBOW W/ TURNING VANES
OR	FLEX DUCT (MAX. LENGTH 4 FT.)
	GATE VALVE
	GLOBE VALVE
	BALL VALVE
	BUTTERFLY VALVE
	CALIBRATED BALANCING VALVE
	PRESSURE REDUCING VALVE (PRV)
	CHECK VALVE
	3-WAY CONTROL VALVE
	STRAINER WITH BLOWDOWN
	CONCENTRIC REDUCER
	ECCENTRIC REDUCER
	VENTURI
	THREADED END FOR HOSE CONNECTION
	BALL VALVE W/ NIPPLE AND CAP
	FLEX CONNECTION
	BLIND FLANGE OR CAP
	PIPING DOWN
	PIPING UP
	PIPING TEE DOWN
S	SOLENOID VALVE
	FLANGE
	UNION
	HIGH EFFICIENCY REFRIGERANT PIPINT TRANSITION
	THERMOMETER
	PRESSURE GAUGE
FS	FLOW SWITCH
	PRESSURE GAUGE W/ SYPHON & COCK
	MANUAL AIR VENT
	PUMP
T	THERMOSTAT
	EQUIPMENT SYMBOL
	DETAIL SYMBOL
M1	SHEET KEYNOTE
	POINT OF CONNECTION
	POINT OF DISCONNECTION
	AIRFLOW DIRECTION
N.O.	NORMALLY OPEN
N.C.	NORMALLY CLOSED
N.I.C.	NOT IN CONTRACT



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DRAWING	BY: MDM	
APPROVED	BY: G.M. J.H. JAC	DATE 6.27.13

PAYETTE SUPERVISOR'S OFFICE

MECHANICAL LEGEND AND SCHEDULES

PROJECT NO. 1555

DRAWING M001

SHEET 75 OF 88

REFRIGERANT FAN COIL SCHEDULE

MARK	MANUFACTURER & MODEL	CFM	COOLING CAP. (MBH)	HEATING CAP. (MBH)	STATIC PRESSURE (IN W.G.)	REFRIGERANT	LIQUID CONNECTION	SUCTION CONNECTION	CONDENSATE CONNECTION	SOUND (DBA)	ELECTRICAL MCA	ELECTRICAL VOLTS/PHASE	DIMENSIONS HxWxD (IN)	WEIGHT (LBS)	REMARKS
FC 1	DAIKIN FXMQ18PVJU	530-636	18.0	20.0	0.8	R-410A	1/4"	1/2"	1 1/4"	41	1.6	208/1	12x40x28	80	WITH ELECTONIC EXPANSION VALVE AND CONDENSATE PUMP
FC 2	DAIKIN FXMQ18PVJU	530-636	18.0	20.0	0.8	R-410A	1/4"	1/2"	1 1/4"	41	1.6	208/1	12x40x28	80	WITH ELECTONIC EXPANSION VALVE AND CONDENSATE PUMP
FC 3	DAIKIN FXMQ12PVJU	247-335	12.0	13.5	0.4	R-410A	1/4"	1/2"	1 1/4"	34	0.6	208/1	12x22x28	55	WITH ELECTONIC EXPANSION VALVE AND CONDENSATE PUMP
FC 4	DAIKIN FXMQ07PVJU	230-317	7.5	8.5	0.4	R-410A	1/4"	1/2"	1 1/4"	33	0.6	208/1	12x22x28	55	WITH ELECTONIC EXPANSION VALVE AND CONDENSATE PUMP
FC 5	DAIKIN FXMQ07PVJU	230-317	7.5	8.5	0.4	R-410A	1/4"	1/2"	1 1/4"	33	0.6	208/1	12x22x28	55	WITH ELECTONIC EXPANSION VALVE AND CONDENSATE PUMP
FC 6	DAIKIN FXMQ18PVJU	530-636	18.0	20.0	0.8	R-410A	1/4"	1/2"	1 1/4"	41	1.6	208/1	12x40x28	80	WITH ELECTONIC EXPANSION VALVE AND CONDENSATE PUMP
FC 7	DAIKIN FXMQ07PVJU	230-317	7.5	8.5	0.4	R-410A	1/4"	1/2"	1 1/4"	33	0.6	208/1	12x22x28	55	WITH ELECTONIC EXPANSION VALVE AND CONDENSATE PUMP
FC 8	DAIKIN FXMQ36PVJU	706-883	30.0	34.0	0.8	R-410A	3/8"	5/8"	1 1/4"	43	2.3	208/1	12x40x28	80	WITH ELECTONIC EXPANSION VALVE AND CONDENSATE PUMP
FC 9	DAIKIN FXMQ07PVJU	230-317	7.5	8.5	0.4	R-410A	1/4"	1/2"	1 1/4"	33	0.6	208/1	12x22x28	55	WITH ELECTONIC EXPANSION VALVE AND CONDENSATE PUMP
FC 10	DAIKIN FXMQ07PVJU	230-317	7.5	8.5	0.4	R-410A	1/4"	1/2"	1 1/4"	33	0.6	208/1	12x22x28	55	WITH ELECTONIC EXPANSION VALVE AND CONDENSATE PUMP
FC 11	DAIKIN FXMQ18PVJU	530-636	18.0	20.0	0.8	R-410A	1/4"	1/2"	1 1/4"	41	1.6	208/1	12x40x28	80	WITH ELECTONIC EXPANSION VALVE AND CONDENSATE PUMP
FC 12	DAIKIN FXMQ12PVJU	247-335	12.0	13.5	0.4	R-410A	1/4"	1/2"	1 1/4"	34	0.6	208/1	12x22x28	55	WITH ELECTONIC EXPANSION VALVE AND CONDENSATE PUMP
FC 13	DAIKIN FXMQ07PVJU	230-317	7.5	8.5	0.4	R-410A	1/4"	1/2"	1 1/4"	33	0.6	208/1	12x22x28	55	WITH ELECTONIC EXPANSION VALVE AND CONDENSATE PUMP
FC 14	DAIKIN FXMQ09PVJU	247-335	9.0	13.5	0.4	R-410A	1/4"	1/2"	1 1/4"	34	0.6	208/1	12x22x28	55	WITH ELECTONIC EXPANSION VALVE AND CONDENSATE PUMP
FC 15	DAIKIN FXMQ09PVJU	247-335	9.0	13.5	0.4	R-410A	1/4"	1/2"	1 1/4"	34	0.6	208/1	12x22x28	55	WITH ELECTONIC EXPANSION VALVE AND CONDENSATE PUMP
FC 16	DAIKIN FXMQ07PVJU	230-317	7.5	8.5	0.4	R-410A	1/4"	1/2"	1 1/4"	33	0.6	208/1	12x22x28	55	WITH ELECTONIC EXPANSION VALVE AND CONDENSATE PUMP
FC 17	DAIKIN FXMQ18PVJU	530-636	18.0	20.0	0.8	R-410A	1/4"	1/2"	1 1/4"	41	1.6	208/1	12x40x28	80	WITH ELECTONIC EXPANSION VALVE AND CONDENSATE PUMP
FC 18	DAIKIN FXMQ12PVJU	247-335	12.0	13.5	0.4	R-410A	1/4"	1/2"	1 1/4"	34	0.6	208/1	12x22x28	55	WITH ELECTONIC EXPANSION VALVE AND CONDENSATE PUMP
FC 19	DAIKIN FXMQ07PVJU	230-317	7.5	8.5	0.4	R-410A	1/4"	1/2"	1 1/4"	33	0.6	208/1	12x22x28	55	WITH ELECTONIC EXPANSION VALVE AND CONDENSATE PUMP
FC 20	DAIKIN FXMQ12PVJU	247-335	12.0	13.5	0.4	R-410A	1/4"	1/2"	1 1/4"	34	0.6	208/1	12x22x28	55	WITH ELECTONIC EXPANSION VALVE AND CONDENSATE PUMP
FC 21	DAIKIN FXMQ18PVJU	530-636	18.0	20.0	0.8	R-410A	1/4"	1/2"	1 1/4"	41	1.6	208/1	12x40x28	80	WITH ELECTONIC EXPANSION VALVE AND CONDENSATE PUMP
FC 22	DAIKIN FXMQ36PVJU	706-883	30.0	34.0	0.8	R-410A	3/8"	5/8"	1 1/4"	43	2.3	208/1	12x40x28	80	WITH ELECTONIC EXPANSION VALVE AND CONDENSATE PUMP
FC 23	DAIKIN FXMQ07PVJU	230-317	7.5	8.5	0.4	R-410A	1/4"	1/2"	1 1/4"	33	0.6	208/1	12x22x28	55	WITH ELECTONIC EXPANSION VALVE AND CONDENSATE PUMP
FC 24	DAIKIN FXMQ96MFVJU	1,200	96.0	59.0	1.0	R-410A	3/8"	7/8"	1"	47	4.1	208/1	19x55x44	271	OUTDOOR AIR PROCESSING UNIT, WITH ELECTRONIC EXPANSION VALVE AND CONDENSATE PUMP

NOTE: ALL FAN COIL UNITS SHALL BE PROVIDED WITH A MERV 13 FILTER.

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DRAWING	BY: G.M., J.H., J.MC	
	CHECK: M.O.M.	
APPROVED: Gina Tant	DISTRICT RANGER	DATE 6-27-13

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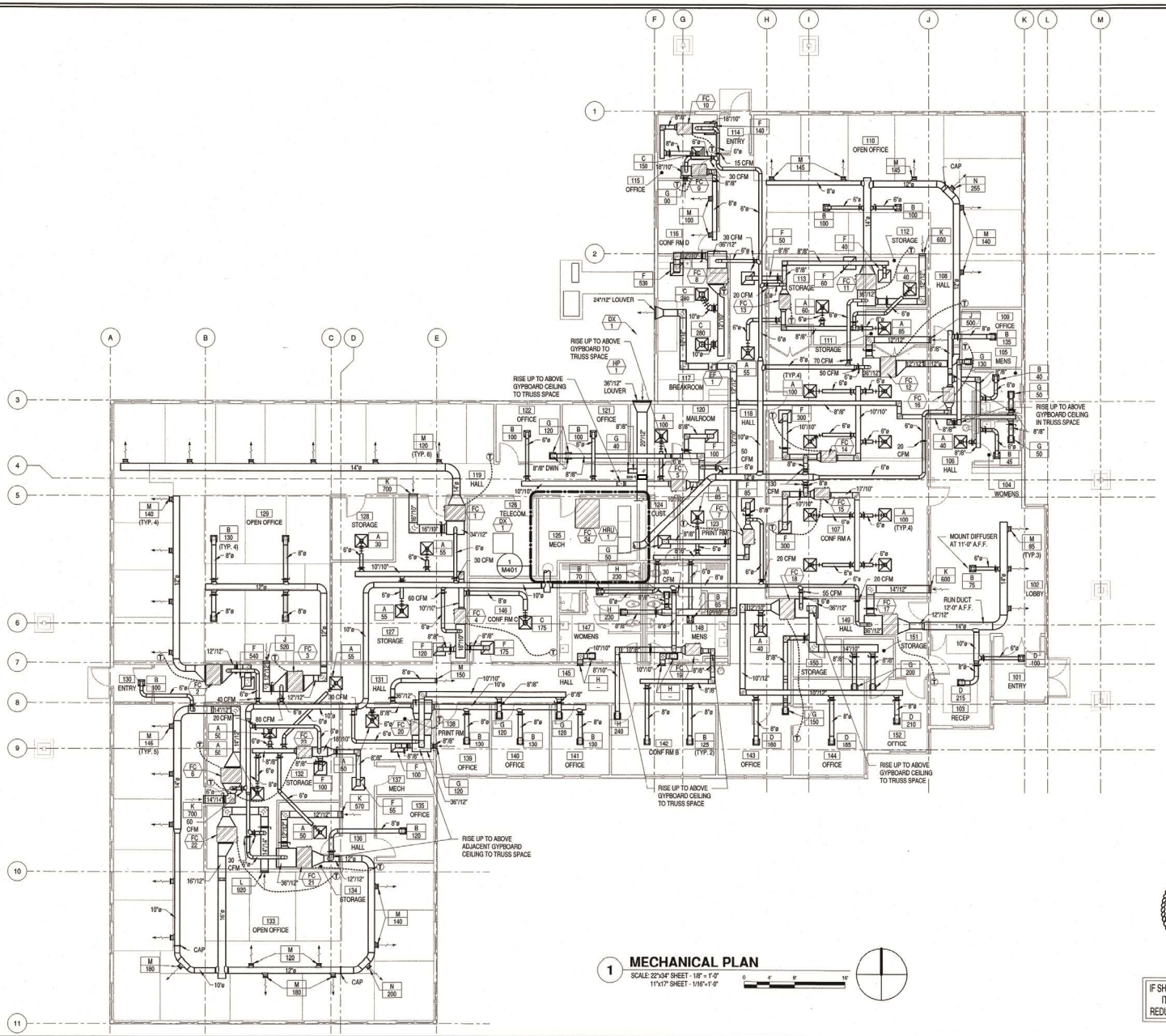
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MECHANICAL SCHEDULES

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PROJECT No.585
DRAWING M002
SHEET 76 OF 88



1 **MECHANICAL PLAN**

SCALE: 22"x34" SHEET - 1/8" = 1'-0"
11"x17" SHEET - 1/16"=1'-0"

0 4' 8' 16'

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APPROVED: Jon Tant DIRECTOR, ENGINEERING		

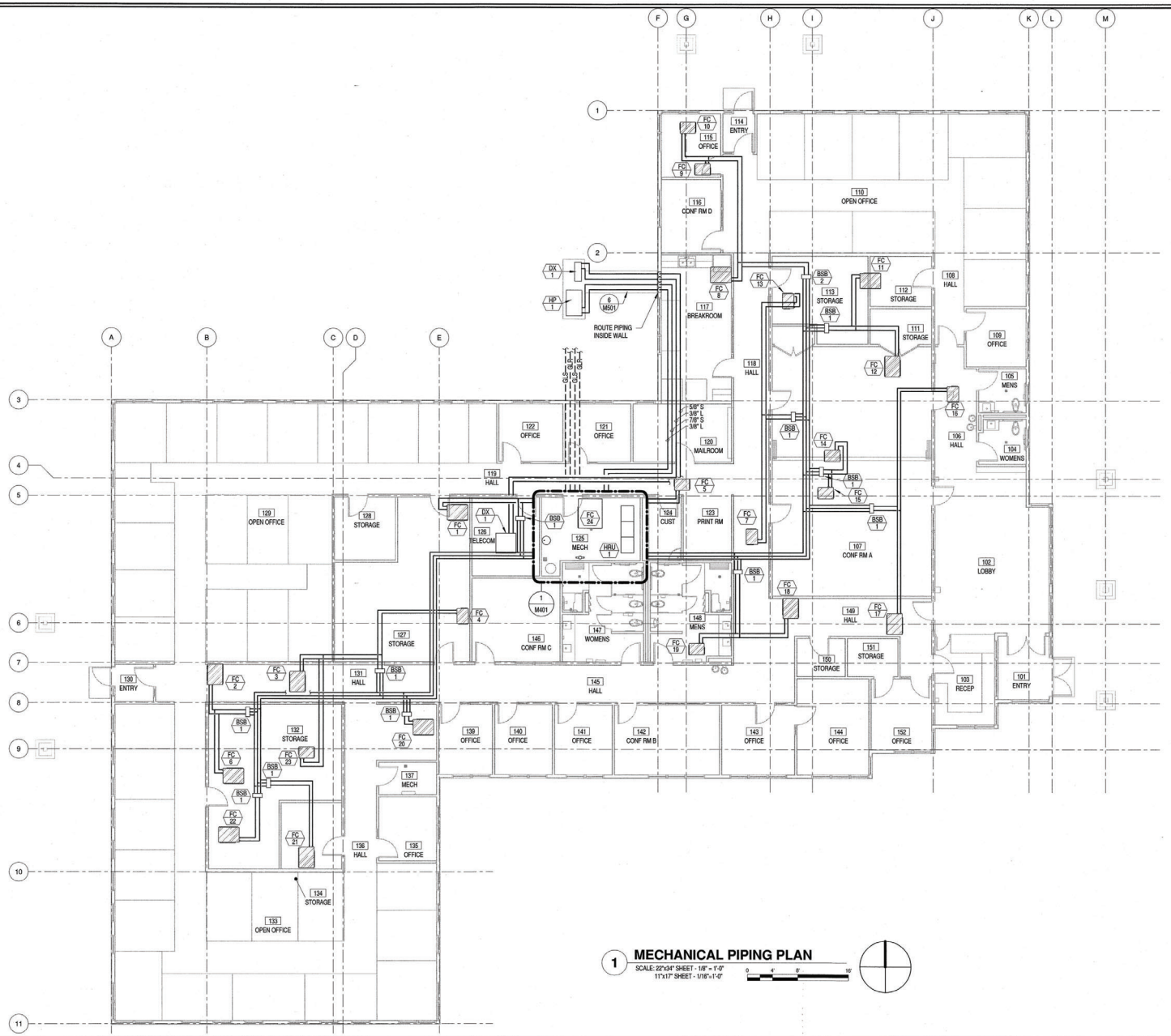
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MECHANICAL PLAN

PROJECT NUMBER

DRAWING M101

SHEET 77 OF 88



1 MECHANICAL PIPING PLAN

SCALE: 22"x34" SHEET - 1/8" = 1'-0"
11"x17" SHEET - 1/16"=1'-0"

0 4' 8' 16'

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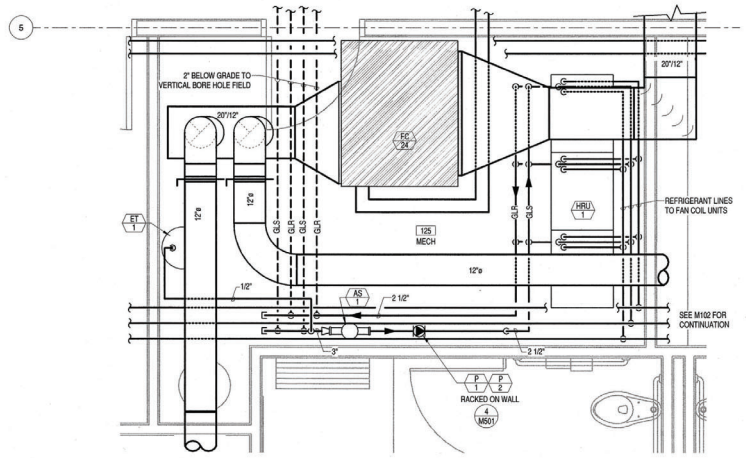
PROJECT No.356

DRAWING M102

SHEET 78 OF 88

PAYETTE SUPERVISOR'S OFFICE

MECHANICAL PIPING PLAN



1 ENLARGED MECHANICAL ROOM PLAN

SCALE: 22"x34" SHEET - 1/2" = 1'-0"
11"x17" SHEET - 1/4"=1'-0"

0 1' 2' 4'

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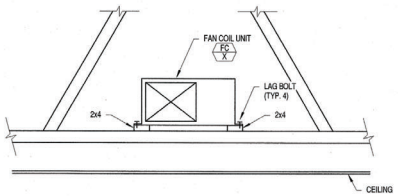
PROJECT NUMBER

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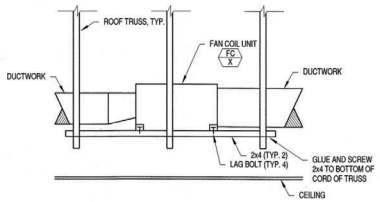
SHEET 79 of 88

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ENLARGED MECHANICAL ROOM PLAN

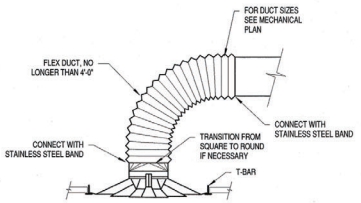


END VIEW

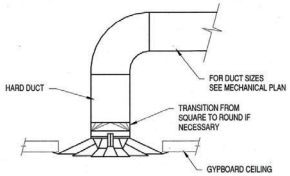


SIDE VIEW

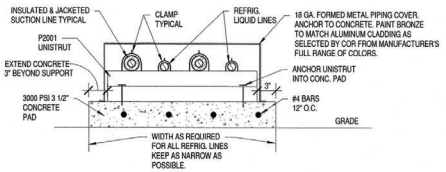
1 FAN COIL UNIT TYPICAL DETAIL
NO SCALE



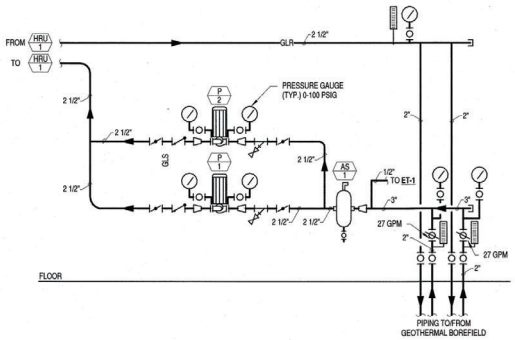
2 DIFFUSER IN LAY-IN CEILING INSTALLATION DETAIL
NO SCALE



3 DIFFUSER IN HARD CEILING INSTALLATION DETAIL
NO SCALE



6 REFRIGERATION PIPING SUPPORT DETAIL
SCALE: NONE



4 INLINE PUMP PIPING DETAIL
NO SCALE

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	DIRECTOR, ENGINEERING	DATE

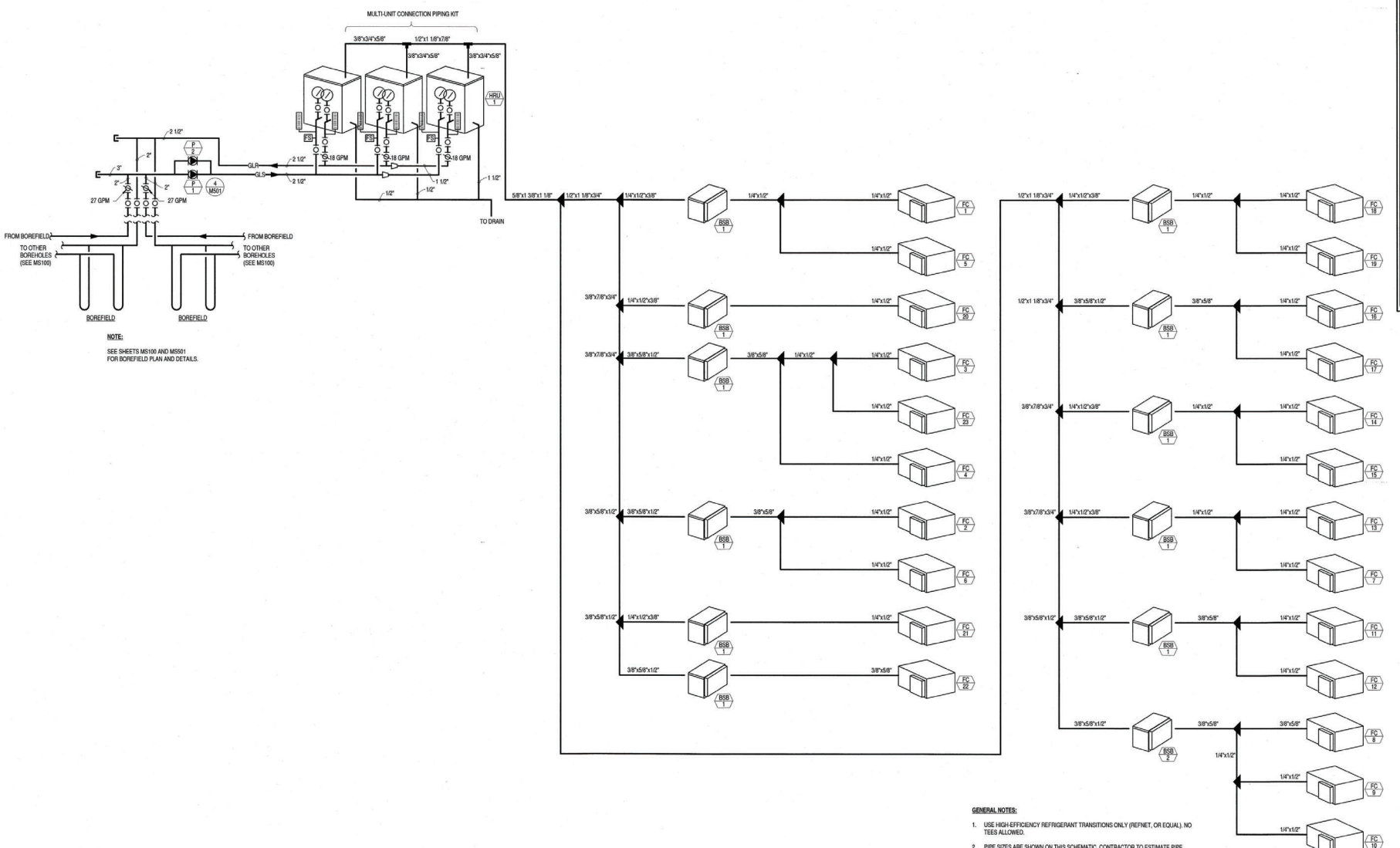
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SHEET 80 OF 88

PAYETTE SUPERVISOR'S OFFICE

MECHANICAL DETAILS



NOTE:

SEE SHEETS MS100 AND MS501
FOR BOREFIELD PLAN AND DETAILS.

GENERAL NOTES:

1. USE HIGH-EFFICIENCY REFRIGERANT TRANSITIONS ONLY (REFNET, OR EQUAL). NO TEES ALLOWED.
2. PIPE SIZES ARE SHOWN ON THIS SCHEMATIC. CONTRACTOR TO ESTIMATE PIPE LENGTHS BASED UPON THE MECHANICAL PIPING PLAN AND THE LOCATION OF EQUIPMENT ON THE PLANS.
3. REFRIGERANT PIPES ARE SHOWN ON THIS SCHEMATIC AS A SINGLE LINE BUT REPRESENT TWO OR THREE PIPES, AS INDICATED BY THE PIPE SIZES LISTED (I.E. 1/4DX1/2D OR 3/8DX5/8DX1/2D). PIPE SIZES INDICATED IN THIS MANNER ARE L X S AND L X HG X S.

1 MECHANICAL PIPING DIAGRAM
NO SCALE

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STATE OF UTAH
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	DISTRICT ENGINEER	

PROJECT NUMBER
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SHEET 81 of 88

PAYETTE SUPERVISOR'S OFFICE

MECHANICAL PIPING DIAGRAM

SEQUENCES OF OPERATION

THE VRF (VARIABLE REFRIGERANT FLOW) HEATING AND COOLING SYSTEM SHALL BE CONTROLLED BY A PROPRIETARY CONTROLS SYSTEM FURNISHED BY THE VRF SYSTEM MANUFACTURER. THIS CONTROLS SYSTEM SHALL CONTROL THE WATER-COOLED HEAT RECOVERY UNIT, GEOTHERMAL LOOP PUMPS, FAN COIL UNITS, BRANCH SELECTOR BOXES, THE OUTDOOR AIR PROCESSING UNIT (SPLIT SYSTEM), THE TELECOMM ROOM UNIT (SPLIT SYSTEM), AND THE RESTROOM EXHAUST FAN.

VRV HEATING AND COOLING SYSTEM

THE FAN COIL UNITS (FC-X) AND ASSOCIATED HEAT RECOVERY UNIT (HRU-1) AND LEAD GEOTHERMAL LOOP PUMP (P-1 OR P-2) SHALL BE STARTED AT THE BEGINNING OF THE OCCUPIED PERIOD (ADJUSTABLE) AND STOPPED AT THE END OF THE OCCUPIED PERIOD. THE VRV CONTROLS SYSTEM SHALL START THE VRV HEATING AND COOLING SYSTEM BASED ON THE TIME SCHEDULE, MONITOR TEMPERATURES IN ALL SPACES SERVED BY THIS SYSTEM, AND OPERATE THE BRANCH SELECTOR BOXES AND ELECTRONIC EXPANSION VALVES ACCORDINGLY TO MAINTAIN THE SPACE TEMPERATURE SETPOINTS. DURING UNOCCUPIED HOURS, THE VRV SYSTEM SHALL RUN INTERMITTENTLY TO MAINTAIN NIGHT SETBACK TEMPERATURE SETPOINTS.

OCCUPIED COOLING SETPOINT: 73 DEG F

UNOCCUPIED COOLING SETPOINT: 80 DEG F

OCCUPIED HEATING SETPOINT: 70 DEG F

UNOCCUPIED HEATING SETPOINT: 60 DEG F

ONE GEOTHERMAL LOOP PUMP SHALL BE THE LEAD PUMP AND THE OTHER THE LAG PUMP. DESIGNATION OF THE LEAD PUMP SHALL BE ALTERNATED MONTHLY. IF THE LEAD PUMP DOES NOT START AT THE BEGINNING OF AN OPERATING PERIOD, OR IF THE LEAD PUMP FAILS DURING OPERATION, THEN THE LAG PUMP SHALL IMMEDIATELY BE STARTED AND OPERATE UNTIL THE END OF THE OPERATING PERIOD.

OUTDOOR AIR PROCESSING UNIT (FC-24) AND ASSOCIATED HEAT PUMP (HP-1)

BASED ON THE TIME OF DAY OPERATING SCHEDULE (ADJUSTABLE), THE OUTDOOR AIR SUPPLY FAN SHALL BE STARTED AND RUN CONTINUOUSLY DURING OCCUPIED HOURS. WHEN THE OUTDOOR AIR TEMPERATURE IS ABOVE 78 DEG F, THE ASSOCIATED HEAT PUMP CONDENSING UNIT (HP-1) SHALL BE STARTED AND THE COMPRESSOR MODULATED TO MAINTAIN A DISCHARGE AIR TEMPERATURE SETPOINT OF 73 DEG F. WHEN THE OUTDOOR AIR TEMPERATURE IS BELOW 55 DEG F, THE HEAT PUMP CONDENSING UNIT (HP-1) SHALL BE STARTED AND THE COMPRESSOR MODULATED TO MAINTAIN A DISCHARGE AIR TEMPERATURE SETPOINT OF 70 DEG F.

RESTROOM EXHAUST FAN (EF-1)

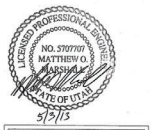
THE RESTROOM EXHAUST FAN SHALL BE STARTED AT THE BEGINNING OF THE OCCUPIED PERIOD (ADJUSTABLE) AND RUN UNTIL THE END OF THE OCCUPIED PERIOD.

TELECOMM ROOM SPLIT SYSTEM (DX-1)

THE COOLING-ONLY SPLIT SYSTEM SERVING THE TELECOMM ROOM SHALL OPERATE CONTINUOUSLY. THE COMPRESSOR SHALL BE MODULATED TO MAINTAIN THE SPACE TEMPERATURE SETPOINT.

ALARMS

IN THE EVENT OF THE FAILURE OF ANY PIECE OF EQUIPMENT CONTROLLED BY THIS SYSTEM, AN ALARM SHALL BE INDICATED AT THE HUMAN-MACHINE INTERFACE SCREEN. ADDITIONALLY, AN ALARM EMAIL OR TEXT MESSAGE SHALL BE SENT TO DESIGNATED DEVICES, AS INPUT BY THE USER/OWNER.



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APPROVED: Steve Tant		DATE 6-27-13
SERVICE ENGINEER		

PROJECT No. 585

PAYETTE SUPERVISOR'S OFFICE

DRAWING M602

SEQUENCE OF OPERATION

SHEET 82 OF 88